

September 2025 Exam

EC400

Probability and Statistical Inference

Suitable for all candidates

Instructions to candidates

This paper contains SEVEN questions. Answer ALL questions.

At the top of your answer scripts, please write the following:

- Your full name;
- Your class group; and
- Your class teacher's name.

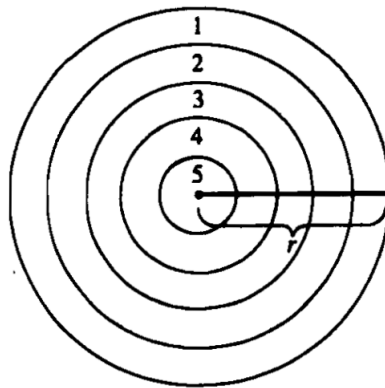
Time Allowed: *2 hours*

You are supplied with: *Standard Normal and Student's t tables*

Calculators: *Calculators are NOT allowed in this examination*

1. **Total: 15 points**

- (a) Let $(\Omega, \mathcal{F}, P)$ represent a probability space with probability measure P . Let the notation “ $\emptyset$ ” denote the null set, and “ $A \setminus B$ ” the set of elements in A but not B for any events $A, B \in \mathcal{F}$.
- State the axioms of probability, and prove that $P(\emptyset) = 0$. **(4pts)**
 - Prove that $P(A \setminus B) = P(A) - P(A \cap B)$. **(3pts)**
- (b) A game of darts is played by throwing a dart at a board and receiving a score corresponding to the number assigned to the region in which the dart lands. Referring to the figure below, we see that the innermost circle is worth 5 points and the outermost ring 1 point. Further, we see that the dart board has radius r and the distance between rings is $r/5$. For a novice player, it seems reasonable to assume that the probability of the dart hitting a particular region is proportional to the area of the region. Thus, a bigger region has a higher probability of being hit. We assume the dart board is always hit on any given throw.



- What is the probability that a novice player will score exactly 1 point on a given throw? **(3pts)**
- Let random variable X denote the number of points scored in a given throw of a dart by a novice player. Explain with detailed reasoning why the probability mass function (PMF) of X is given by:

$$P(X = i) = \frac{(6 - i)^2 - (5 - i)^2}{5^2},$$

for $i = 1, \dots, 5$, and 0 otherwise. Argue that $P(X = i)$ is a valid PMF independently of A , where $A = \pi r^2$, is the size of the dart board. **(5pts)**

The answer for part (a)i is:

For events $A_1, A_2, \dots \in \mathcal{F}$,

Axiom 1. $\Pr(A_i) \geq 0$ for $i = 1, 2, \dots$; **1 point**.

Axiom 2. $\Pr(\Omega) = 1$; and **1 point**.

Axiom 3. For any events A_i and A_j , where $i \neq j$, such that $A_i \cap A_j = \emptyset$, it holds that

$$\Pr\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} \Pr(A_i).$$

1 point. Noting that $\emptyset \cup \Omega = \Omega$ constitutes a union of disjoint events, we can use Axioms 2 and 3 to prove the required result that $P(\emptyset) = 1 - 1 = 0$. **1 point**.

The answer for part (a)ii is:

Notice that $(A \cap B) \cup (A \cap B^c) = A$ constitutes a union of disjoint events. **1 point**.

By Axiom 3 **1 point**. and noticing that $P(A \cap B^c)$ is nothing other than $P(A \setminus B)$ **1 point**. the required result follows.

The answer for part (b)i is:

$$\frac{\pi r^2 - \pi(4r/5)^2}{\pi r^2} = 1 - (4/5)^2$$

Upto 3 points for this or any equivalent method/answer.

The answer for part (b)ii is:

There are 5 concentric circles depicted with (in decreasing size order) radii $(6 - i)r/5$ for $i = 1, 2, 3, 4, 5$. The first 4 scoring regions are accordingly defined by the difference in area between a disc with radius $(6 - i)r/5$ and a smaller disc with radius $(6 - (i + 1))r/5$ for $i = 1, \dots, 4$. The fifth and final scoring region is simply the innermost disc, i.e. with radius $r/5$. We thus obtain

$$P(X = i) = \frac{(\pi((6 - i)r/5)^2 - \pi((5 - i)r/5)^2)}{\pi r^2},$$

as the i^{th} scoring region divided by the overall available surface area of the dart board.

Upto 3 points for this or any equivalent method/answer. For instance, students can frame the answer in terms of CDFs and then use $F_X(i + 1) - F_X(i)$ to recover the PMF. For validity, students should argue that the PMF is everywhere non-negative and that it adds to 1 **1 point**. and this is true irrespective of the size of the dartboard because the radius, r , just drops out of the ratio given above (and also π). **1 point**.

2. Total: 15 points

Long Shorts, Inc., has two trading desks, A and B . Desk A executes $2/3$ of all trades, and desk B executes $1/3$. The probability that a trade results in a loss differs by desk: 5% of trades from A result in a loss, while 10% of trades from B result in a loss.

- (a) Let L denote the event that a randomly chosen trade results in a loss. Compute $P(L)$ using the law of total probability. **(3pts)**
 - (b) A trade is chosen at random and is found to have resulted in a loss. Compute the probability that it was executed by desk B , i.e. find $P(B | L)$. **(4pts)**
 - (c) Consider the two-desk system in general terms: $P(A) = p$, $P(B) = 1 - p$, $P(L | A) = \alpha$, $P(L | B) = \beta$, with $0 < p, \alpha, \beta < 1$. Show algebraically that if $\beta > \alpha$, then $P(B | L) > P(B)$, and explain why knowing that a trade resulted in a loss raises our belief that it was executed by desk B (relative to the unconditional probability that it was executed by desk B). **(5pts)**
 - (d) Suppose *Long Shorts, Inc.* introduces a risk audit that only examines trades that resulted in losses. Let R denote the event that a trade is selected for audit. Do you agree or disagree with the claim that $P(B | R) = P(B | L)$? Explain intuitively. (Algebra is not necessary for full marks.) **(3pts)**
-

(a) Law of total probability

$$P(L) = P(A)P(L|A) + P(B)P(L|B) = \frac{2}{3} \cdot \frac{1}{20} + \frac{1}{3} \cdot \frac{1}{10} = \frac{2}{60} + \frac{2}{60} = \frac{4}{60} = \frac{1}{15} \text{ (3 pts - give partial credit if appropriate)}$$

(b) Bayes' theorem

$$P(B | L) = \frac{P(B)P(L|B)}{P(L)} = \frac{\frac{1}{3} \cdot \frac{1}{10}}{\frac{1}{15}} = \frac{1/30}{1/15} = \frac{1}{2} \text{ (4 pts - give partial credit if appropriate)}$$

(c) Conditional probability comparison

Algebra

$$P(B | L) = \frac{(1-p)\beta}{p\alpha + (1-p)\beta} \text{ so let's compare to } P(B) = 1 - p:$$

$$P(B | L) > 1 - p \iff$$

$$(1-p)\beta > (1-p)(p\alpha + (1-p)\beta) \iff$$

$$(1-p)\beta - (1-p)^2\beta > (1-p)p\alpha \iff$$

$$(1-p)(\beta - \beta + \beta p) > (1-p)p\alpha \iff \beta > \alpha \text{ (3 pts - give partial credit if appropriate)}$$

Intuition

Desk B has a higher loss rate ($\beta > \alpha$), so observing a loss increases the probability the trade came from B compared with the unconditional probability (which is our

prior belief in the absence of any information on whether the trade was a loss or not).

(2 pts - give partial credit if appropriate)

(d) Risk audit / conditional probability

Agree. Knowing that a trade was audited is the same as knowing it was a loss, since only losses are ever audited. The audit procedure does not lean toward losses from desk A or desk B ; every loss is equally eligible. Therefore, conditioning on R gives no extra information beyond conditioning on L . This is why $P(B | R) = P(B | L)$ even though $R \subseteq L$. (3 pts - give partial credit if appropriate)

3. Total: 15 points

Let $X \sim \text{Exponential}(\lambda)$ with rate parameter $\lambda > 0$. In other words, the probability density function (PDF) of X is given by:

$$f_X(x) = \begin{cases} \lambda e^{-\lambda x}, & x > 0 \\ 0, & \text{otherwise.} \end{cases}$$

- (a) Derive the moment-generating function (MGF) of X , say $M_X(t)$, and state the values of t for which $M_X(t)$ is well-defined. (3pts)
 - (b) Using the MGF, obtain the expected value and variance of X . (3pts)
 - (c) Derive the cumulative distribution function (CDF) of X . (3pts)
 - (d) Let $Z \sim \text{Bernoulli}(p)$. Find the expected value and variance of Z . (3pts)
 - (e) Now let $Y = \mathbb{1}(X \leq c)$ where c is a positive constant, and $\mathbb{1}(\cdot)$ is the indicator function. Find the expected value and variance of Y . (3pts)
-

For part (a):

$$M_X(t) = \frac{\lambda}{\lambda - t} \text{ for } t < \lambda.$$

1 point for direct/indirect evidence that students know what is an MGF. 1 point for final expression. 1 point for range of t .

For part (b):

$$E(X) = \frac{1}{\lambda} \text{ and } \text{Var}(X) = \frac{1}{\lambda^2}.$$

1 point for expected value and 2 points for variance.

For part (c):

$$F_X(x) = \mathbb{1}(x \leq 0) \times 0 + \mathbb{1}(x > 0) \times (1 - e^{-\lambda x}), \text{ where we use the usual notation for}$$

an indicator function.

1 point for precisely specified support and 2 points for the integral itself

For part (d):

$$E(Z) = p \text{ and } Var(Z) = p(1 - p)$$

1 point for any direct/indirect evidence that students know the Bernoulli PMF. 1 point for expected value and 1 point for variance.

$$E(Y) = 1 - e^{-\lambda c} \text{ and } Var(Y) = e^{-\lambda c}(1 - e^{-\lambda c})$$

1 point for any direct/indirect evidence that students recognised that

$$Y \sim \text{Bern}(1 - e^{-\lambda c}).$$

1 point for expected value and 1 point for variance.

4. Total: 15 points

Suppose the random variable X has probability density function (PDF) given by:

$$f_X(x) = \begin{cases} 2x, & 0 \leq x \leq 1, \\ 0, & \text{otherwise.} \end{cases}$$

Let $Y = \sqrt{X}$ and define θ to be the fourth (raw) moment of Y .

(a) Find the PDF of Y using the change-of-variable formula. **(4pts)**

(b) Suppose we have a single observation: X . Consider the estimator

$$\hat{\theta} = Y^4.$$

Show that the mean squared error (MSE) of $\hat{\theta}$ is $1/12$. **(4pts)**

(c) Now suppose we have a random sample of size n : $X_1, \dots, X_n$. Consider the estimator

$$\hat{\theta}_n = \frac{1}{n} \sum_{i=1}^n (Y_i^4 + k),$$

where $k \in \mathbb{R}$ is a fixed constant. Compute the MSE of $\hat{\theta}_n$ in terms of its bias, the sample size n , and $Var(\hat{\theta})$. **(4pts)**

(d) Discuss the behaviour of the MSE of $\hat{\theta}_n$ as $n \rightarrow \infty$. Under what conditions on k is $\hat{\theta}_n$ consistent? **(3pts)**

-
- (a) PDF of Y : Let $Y = \sqrt{X} \implies X = Y^2, \frac{dX}{dY} = 2Y$. Then

$$f_Y(y) = f_X(y^2) \left| \frac{dx}{dy} \right| = 2(y^2) \cdot 2y = 4y^3, \quad 0 \leq y \leq 1.$$

1 point for formula, 1 point for calculations, 1 point for PDF, 1 point for support.

- (b) Single-observation estimator $\hat{\theta} = Y^4$: Using $f_Y(y)$:

$$\begin{aligned} E[\hat{\theta}] &= E[Y^4] = \int_0^1 y^4 \cdot 4y^3 dy = \int_0^1 4y^7 dy = \frac{1}{2} \implies \text{Bias} = 0 \\ E[Y^8] &= \int_0^1 y^8 \cdot 4y^3 dy = \int_0^1 4y^{11} dy = \frac{1}{3} \implies \text{Var}(\hat{\theta}) = E[Y^8] - (E[Y^4])^2 = \frac{1}{12} \\ \text{MSE}(\hat{\theta}) &= \text{Var} + \text{Bias}^2 = \frac{1}{12} \end{aligned}$$

1 point for bias, 2 points for variance, 1 point for MSE. (Note that students can also use the original PDF of X to arrive at the same answer for moments of Y .)

- (c) Sample mean estimator $\hat{\theta}_n = \frac{1}{n} \sum (Y_i^4 + k)$:

$$\begin{aligned} \text{Bias}(\hat{\theta}_n) &= k, \quad \text{Var}(\hat{\theta}_n) = \frac{1}{n} \text{Var}(Y^4) = \frac{1}{12n} \\ \text{MSE}(\hat{\theta}_n) &= \text{Var} + \text{Bias}^2 = \frac{1}{12n} + k^2 \end{aligned}$$

1 point for bias, 2 points for variance, 1 point for MSE.

- (d) Asymptotic behavior: As $n \rightarrow \infty$, $\text{Var}(\hat{\theta}_n) \rightarrow 0$, so $\text{MSE} \rightarrow k^2$. 1 point.
- If $k = 0$, $\text{MSE} \rightarrow 0$, estimator is consistent. 1 point.
- If $k \neq 0$, bias dominates, $\text{MSE} \not\rightarrow 0$, estimator is inconsistent. 1 point.
-

5. Total: 10 points

Suppose $X_i \stackrel{iid}{\sim} \text{Geometric}(\pi)$ for $i = 1, \dots, n$, where π is unknown. In other words, the probability mass function of each of the X_i variables is given by

$$\Pr(X_i = x) = \begin{cases} (1 - \pi)^{x-1} \pi & x = 1, 2, \dots \\ 0 & \text{otherwise.} \end{cases}$$

Derive the maximum likelihood estimator for π given a sample $\mathbf{x} = (x_1, \dots, x_n)'$.

(You do **not** need to verify the second-order condition.)

(10pts)

Q5. The likelihood function is

$$\mathcal{L}(\pi; x) = P(X_1 = x_1, \dots, X_n = x_n; \pi) \stackrel{\text{IID}}{=} \prod_{i=1}^n P(X_i = x_i)$$

$$= \prod_{i=1}^n (1-\pi)^{x_i-1} \pi = (1-\pi)^{\left(\sum_{i=1}^n x_i - n\right)} \pi^n$$

where $X := (X_1, \dots, X_n)'$.

The log-likelihood function is

$$\ell(\pi; x) = \log \mathcal{L}(\pi; x) = \left[\sum_{i=1}^n x_i - n \right] \log(1-\pi) + n \log \pi$$

The score function is

$$\mathcal{A}(\pi; x) = \left(\frac{\sum_{i=1}^n x_i - n}{1-\pi} \right) + \frac{n}{\pi}.$$

Solving the equation $\mathcal{A}(\hat{\pi}_{MLE}; x) = 0$ gives us $\hat{\pi}_{MLE} = 1/\bar{x}$.

2 points for likelihood; 2 for log-likelihood; 2 for first derivative; 2 for final answer; award the final 2 marks if student remembers to explicitly put a hat on π when setting score to zero.

6. **Total: 20 points**

- (a) i. Congratulations! You have been hired to teach a class on hypothesis testing for BSc Economics students at LSE. You are preparing your class notes, in which you need to list the key steps involved in carrying out a hypothesis test. What do you write? (Provide 1-2 lines of explanation for each step.) **(5pts)**

- ii. Your students love your teaching notes! However, they completely forget your guidance when they do their end-of-term empirical research papers. For instance, here is an excerpt from a randomly chosen student's paper:

*We want to see if **BrainJuice** increases IQ. We measured 16 people's IQ before and after drinking **BrainJuice**. The mean IQ before was 110, after was 115. We assumed normality, ran a one-sided test, and got $p = 0.02$. In other words, there is only a 2% chance that the null is true. Therefore, we conclude that **BrainJuice** increases IQ.*

Sigh! Identify at least 5 distinct mistakes and/or ambiguities in this student's work. Provide the student with constructive feedback in each case.

(5pts)

- iii. As you flip through the various research papers, you find a much better submission, from which the following extract immediately catches your attention:

Suppose $H_0 : \theta = 0$ and $H_1 : \theta \neq 0$. Power is defined as:

$$P(\text{Reject } H_0) = P(\text{Reject } H_0 | H_0 \text{ is true})P(H_0 \text{ is true}) \\ + P(\text{Reject } H_0 | H_0 \text{ is false})P(H_0 \text{ is false})$$

Comment on the above extract. Is the student's algebra correct? If not, identify the mistake(s), write down the correct algebraic definition of the power of a test, and explain why the student's analysis might be misleading.

(5pts)

- (b) The random variable X is normally distributed with mean μ and variance $\sigma^2 = 4$. A sample of $n = 16$ independent realisations of X was collected and the sample mean was $\bar{x} = 20.5$. Find a 90% confidence interval for μ . (You may state your answers as simplified expressions rather than exact numerical values.) **(5pts)**

For part (a)i:

- **Assumptions** (arguably, the most critical step since it affects many further steps below)
- **Hypotheses** (clear statement of the null, H_0 , and the alternative)
- **Test Statistic, say V** (function of the data that holds information value for H_0)
- **Distribution of V under H_0 and assumptions** (usually one of $N(0, 1)$, t , χ^2 , or F)
- **Significance Level** (some exogenously chosen $\alpha \in (0, 1)$, typically small like 0.05 or 0.01)
- **Critical Value** (usually from a statistical table)
- **Rejection Rule** (demarcate a region in the domain of the PDF of V whereby we reject H_0)
- **Computed value of V** (in order to be able to implement the test)
- **Test Result** (compare findings against our rejection rule and make a decision¹)
- **Interpretation** (framing the result back in terms of the underlying context)

Award 0.5 points for each of the 10 steps.

For part (a)ii, there are many possible answers. I request the TAs to please use their judgement in marking this question. Types of responses could include (but are not limited to):

- The entire paragraph is just sloppy. For instance, rather humorously, the language is so bad that it isn't even clear whether the researchers were drinking BrainJuice themselves or actually gave it to their subjects (see 2nd sentence)!
- The assumptions aren't clearly laid out. They mention normality but what is the random variable they are talking about? Is the variance assumed known or unknown? Independence? What exactly is the model specification and the context for the hypothesis test?

¹You never accept the null. You either reject or fail to reject H_0 .

Further, you must clarify at exactly what significance level your result holds. Either that, or directly report the p -value.

- The null is not explicitly mentioned. Neither is the alternative – in fact, it isn't even clear whether it is a left-tailed or right-tailed one-sided test!
- The test statistic isn't given. We don't know the distribution under the null either since the researcher hasn't mentioned whether the variance is known or not. Did they use a t or standard normal distribution?
- There is no clear phrasing to indicate that the student understands the distinction between population and sample means.
- There is a glaring error in the interpretation of the p value. It is NOT the probability that the null is true. It is the probability (assuming the null is true) that we find a value for our test statistic equal to or more extreme (adverse to the null) than was obtained in our sample. Alternatively, it is the minimum level of significance at which the null is rejected.
- No caveats are provided. For instance, how was treatment assigned? Was it a pairwise comparison? If not, was it randomised over the sample? If not, are there any confounders to worry about? Is the sample size enough? Is normality a justifiable assumption? How was IQ measured?
- You cannot accept the null; you can only fail to reject it. The conclusion is too strong (bordering on ignorant/arrogant).

Award upto 5 points for any answer that captures the gist of the arguments I have made above. (Do not give any marks if they just say lazy things like “see part (a)i”! That isn't really constructive feedback.)

For part (a)iii: This is much more interesting. There is no sloppiness in that the algebra and the notation is all precise. Indeed, the formula the student has invoked is clearly the law of total probability. 1 point for recognising the latter. Rather, the mistakes are in (1) the definition of power of a test and (2) the conceptual notion of what is a parameter and what is a random variable! See below:

- I suspect most EC400 students will answer here that the definition of power is incorrect. That is, power is not the unconditional probability that the null is rejected. Rather, it is the conditional probability of rejecting the null given that the null is false (measured as a function of the parameter value under consideration). 2 points for clarifying power

- I suspect fewer EC400 students will catch here that the student is clearly confused about the fact that a parameter is not a random variable (at least not in frequentist statistics). The student is applying the law of total probability somewhat robotically without thinking through what they are saying. There's no concept of a 'probability that null is true' or '...null is false' in classical statistics; that is, it's very silly to talk about things like $P(\theta = 0)$. The parameter doesn't jump around in different states of the world - it is not a stochastic entity. It is simply a fixed unknown feature of the model that we maintain as governing the probabilistic behaviour of a random variable. **2 points for recognising this.**

For part (b): This is almost exactly like one of the problem set questions. I've just changed the confidence level. Students should give the final answer:

$$[20.5 - (1/2)1.645, 20.5 + (1/2)1.645].$$

2 points for 1.645. 1 point for standard error (1/2). 2 points for final answer - accurately expressed.

7. **Total: 10 points**

(a) Give a precise definition for: (3pts)

- i. a χ_n^2 random variable;
- ii. a t_ν random variable; and
- iii. an $F_{p,q}$ random variable.

(b) Suppose $Z \sim \text{Normal}(0, 1)$ and define $U = Z^2$. Consider the ratio

$$T = \frac{Z}{\sqrt{U/1}}.$$

Does T follow the t distribution? Explain your reasoning. (3pts)

(c) Suppose $U \sim \chi_k^2$ and $V \sim \chi_m^2$ are independent. Consider the ratio

$$W = \frac{U/m}{V/k}.$$

Does W follow the F distribution? If so, specify the degrees of freedom; if not, explain how its stochastic behaviour is related to the F -distribution. (4pts)

(a) Precise definitions: 1 point each

- i. $X \sim \chi_n^2$ iff there exist iid $Z_1, \dots, Z_n \sim N(0, 1)$ such that $X = \sum_{i=1}^n Z_i^2$.
- ii. $T \sim t_\nu$ iff $T = \frac{Z}{\sqrt{W/\nu}}$, where $Z \sim N(0, 1)$, $W \sim \chi_\nu^2$, and Z indep. of W .
- iii. $F \sim F_{p,q}$ iff $F = \frac{X/p}{Y/q}$, where $X \sim \chi_p^2$, $Y \sim \chi_q^2$ are independent.

(b) Suppose $Z \sim N(0, 1)$ and $U = Z^2$. Consider $T = \frac{Z}{\sqrt{U/1}}$. Then $T = \frac{Z}{\sqrt{Z^2}} = \frac{Z}{|Z|} = \begin{cases} +1, & Z > 0 \\ -1, & Z < 0 \end{cases}$ with probability 1/2 each. Hence T is discrete, not a t -variable, because the denominator depends on the numerator, violating the independence requirement.
upto 3 points depending on quality of answer.

(c) Suppose $U \sim \chi_k^2$ and $V \sim \chi_m^2$ are independent. The standard $F_{k,m}$ distribution is $F = \frac{U/k}{V/m} \sim F_{k,m}$. For the given ratio $W = \frac{U/m}{V/k} = \frac{k^2}{m^2} F_{k,m}$. Thus W is not exactly $F_{k,m}$ unless $k = m$; in general, W is a constant multiple of an $F_{k,m}$ variable. 2 points to recognise not F . 2 points for additional algebraic details.

END OF EXAM PAPER

TABLE FD.1 Observations on Income and Education, 20 Observations

I = Observation,
 Y = Income.

Source: Data are artificial.

APPENDIX G



STATISTICAL TABLES

TABLE G.1 Cumulative Normal Distribution. Table Entry Is $\Phi(z) = \text{Prob}[Z \leq z]$

[illegible]

TABLE G.2 Percentiles of the Student's t Distribution. Table Entry Is x Such that $\text{Prob}[t_n \leq x] = P$

n	.750	.900	.950	.975	.990	.995
1	1.000	3.078	6.314	12.706	31.821	63.657
2	.816	1.886	2.920	4.303	6.965	9.925
3	.765	1.638	2.353	3.182	4.541	5.841
4	.741	1.533	2.132	2.776	3.747	4.604
5	.727	1.476	2.015	2.571	3.365	4.032
6	.718	1.440	1.943	2.447	3.143	3.707
7	.711	1.415	1.895	2.365	2.998	3.499
8	.706	1.397	1.860	2.306	2.896	3.355
9	.703	1.383	1.833	2.262	2.821	3.250
10	.700	1.372	1.812	2.228	2.764	3.169
11	.697	1.363	1.796	2.201	2.718	3.106
12	.695	1.356	1.782	2.179	2.681	3.055
13	.694	1.350	1.771	2.160	2.650	3.012
14	.692	1.345	1.761	2.145	2.624	2.977
15	.691	1.341	1.753	2.131	2.602	2.947
16	.690	1.337	1.746	2.120	2.583	2.921
17	.689	1.333	1.740	2.110	2.567	2.898
18	.688	1.330	1.734	2.101	2.552	2.878
19	.688	1.328	1.729	2.093	2.539	2.861
20	.687	1.325	1.725	2.086	2.528	2.845
21	.686	1.323	1.721	2.080	2.518	2.831
22	.686	1.321	1.717	2.074	2.508	2.819
23	.685	1.319	1.714	2.069	2.500	2.807
24	.685	1.318	1.711	2.064	2.492	2.797
25	.684	1.316	1.708	2.060	2.485	2.787
26	.684	1.315	1.706	2.056	2.479	2.779
27	.684	1.314	1.703	2.052	2.473	2.771
28	.683	1.313	1.701	2.048	2.467	2.763
29	.683	1.311	1.699	2.045	2.462	2.756
30	.683	1.310	1.697	2.042	2.457	2.750
35	.682	1.306	1.690	2.030	2.438	2.724
40	.681	1.303	1.684	2.021	2.423	2.704
45	.680	1.301	1.679	2.014	2.412	2.690
50	.679	1.299	1.676	2.009	2.403	2.678
60	.679	1.296	1.671	2.000	2.390	2.660
70	.678	1.294	1.667	1.994	2.381	2.648
80	.678	1.292	1.664	1.990	2.374	2.639
90	.677	1.291	1.662	1.987	2.368	2.632
100	.677	1.290	1.660	1.984	2.364	2.626
∞	.674	1.282	1.645	1.960	2.326	2.576